RESEARCH DESIGN

Japanese Learning Program for first-year M30 students

Research question

How do app-based self-study and instructor-led learning compare in their effectiveness for vocabulary acquisition among Minervan beginner Japanese learners when keeping study time and content constant?

Hypotheses

Focal hypothesis

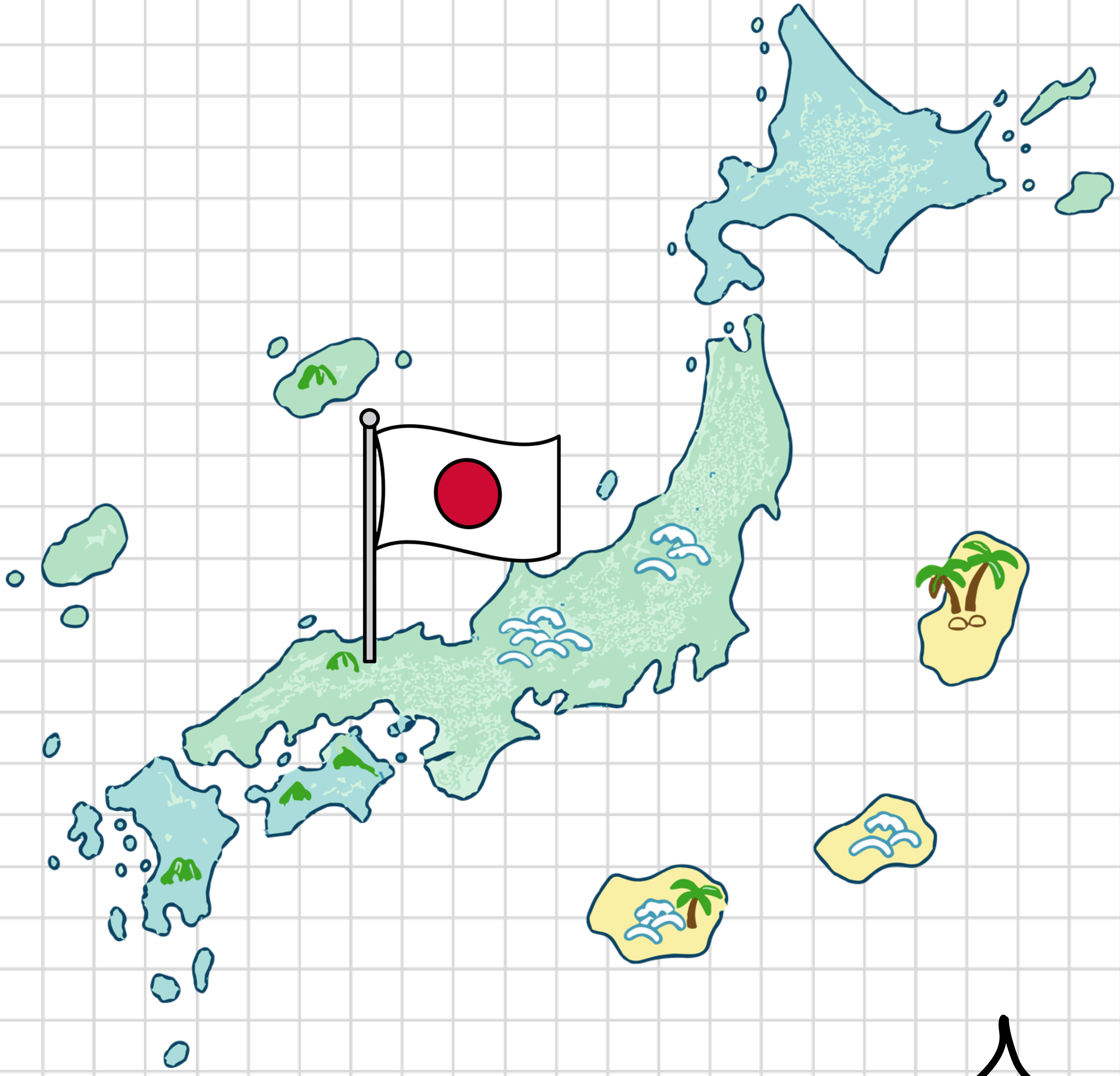
If students learn vocabulary through instructor-led interactive learning, then they will achieve higher vocabulary acquisition, because interactive dialogue promotes deeper cognitive processing of language.

Alternative hypothesis

If students learn vocabulary through app-based learning, then they will achieve higher vocabulary acquisition, because algorithm-based review strengthens memory through repeated retrieval.

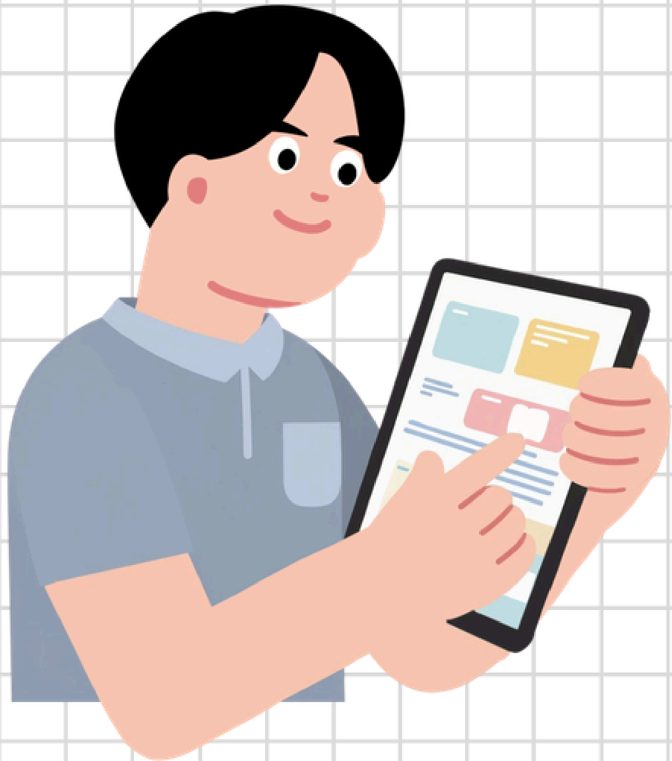
Background

- Minerva students relocate from San Francisco to Tokyo in Year 2
- Basic Japanese is needed for daily life and community interaction



Learning methods

Instructor-led learning



App-based learning

Study design

Randomized controlled trial

(Interventional study)

- Reduce confounding variables' effects
- Investigate causation

Duration: 8 weeks

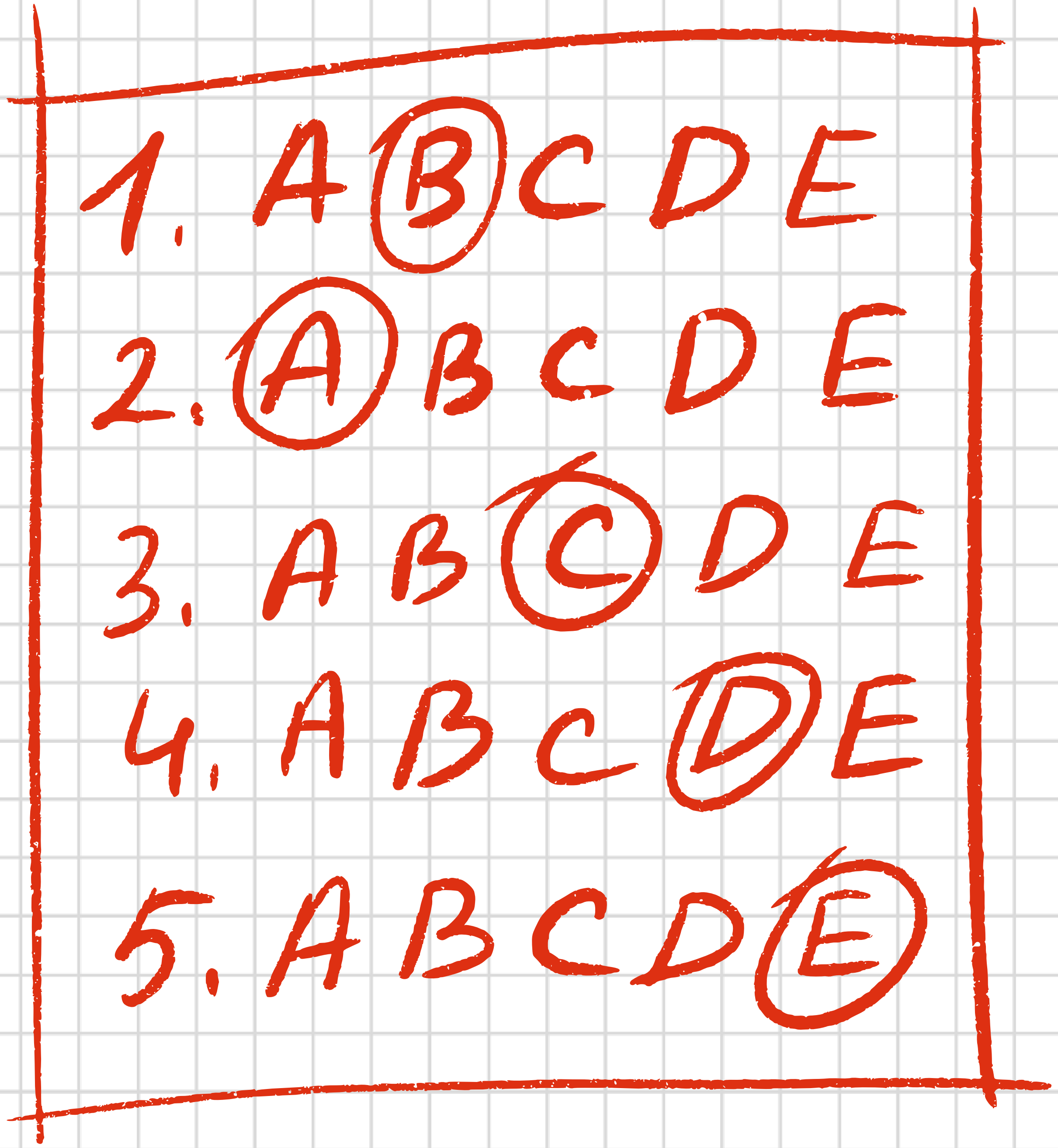
Predictions

The average vocabulary test score of students in the instructor-led learning condition will be higher than the average score of students in the app-based condition at the end of the eight-week program.

The average vocabulary test score of students in the app-based condition will be higher than that of students in the instructor-led condition.

There is no significant differences in average scores between two groups

Measuring method



Variables

Independent variables

Learning format

- App-based self-study
- Instructor-led in-person learning

Dependent variables

Vocabulary acquisition score

Confounding variables

- Language aptitude
- Prior Japanese exposure
- Personal preference
- Curriculum effect
- Motivation

Data collection

Outcomes

Vocabulary acquisition

Data source

Standardized vocabulary tests

Timeline

0 week

8 weeks

Learning program

Test

Sampling strategy

Population	Eligibility	Sampling approach
First-year M30s	No formal Japanese study	Convenience sampling

Comparison groups

Participants (M30s)

Baseline survey (motivation)

Block randomization
Pair students with similar motivation scores

Randomization
- Group A: app-based
- Group B: instructor-leb

Assumptions

Participants follow the assigned learning method and study schedule

Eight-week learning period is sufficient to observe measurable differences in vocabulary learning between two methods

Vocabulary tests reflect learning outcomes

Participants do not use the other learning method

Strengths

Randomized controlled trial design

Confounders control

Bias mitigation

Weaknesses

Small sample size

Artificial learning environment

Limited external validity

Conclusion

Expected results

Instructor-led learning
App-based learning

Impact

The study focuses on learning efficiency to identify potential causation

Further research

Further research can include larger student populations, longer learning periods

THANK YOU FOR LISTENING